

ZHANG Ziyang Ph.D.

AI Systems | AI for Science and Engineering

z_zz@u.nus.edu | +65-89428966



PROFESSIONAL EXPERIENCE

Consultant Specialist HSBC | Enterprise AI & FinTech

July 2026–Present

Founder & AI Lead Jue-Ming Technology | AI/LLM Software Development

April 2025–June 2026

EDUCATION

Ph.D. Biomedical Engineering, National University of Singapore.

Aug 2020–Dec 2024

- Joint Ph.D. training with Pharmacy and Pharmaceutical Sciences. (2022–2024)
- Joint Ph.D. training with Temasek Life Sciences Laboratory and National Cancer Centre Singapore. (2020–2022)
- 3D Bio-Printing and AI-assisted Biomaterial Discovery for Diabetic Wound Care. Thesis

B.Sc. Chemistry, Huazhong University of Science and Technology.

Aug 2016–Jun 2020

SELECTED PROJECTS

Enterprise AI Evaluation & Capability Lifecycle

Core Contributor | 2026–Present

- Lead development of telemetry and evaluation workflows for enterprise AI capabilities, spanning usage measurement, telemetry integration, dashboarding, capability assessment, and end-to-end validation.
- Extend shared platform components and co-develop capability-evaluation methods with other engineers, driving iterative integration, testing, defect resolution, and governed capability delivery.

AI Solutions Development & Delivery

Algorithm Lead | 2025–2026

- Chemical QA: Applied computer vision to particle images to quantify particle-size distributions and diffusion statistics.
- Web crawl: indexed and incrementally maintained a WeChat Official Account corpus (~800k Chinese characters).
- App development: UI prototyping; prompt engineering with RAG; red-teaming and guardrails for attack interception.
- LLM fine-tuning: built repository-grounded training-data workflows; fine-tuned Qwen2.5 with PEFT/QLoRA (Unsloth/Hugging Face), evaluated inference, and packaged the model for deployment.

AI Molecule Generation – Cheminformatics Evaluation

Simulation & Evaluation | 2024–2026

- Used Boltz-2 to predict small-molecule–protein complexes for cross-validation against the broader workflow.
- Used AlphaFold to model small-molecule–protein (Amlexanox–IKK ϵ), peptide–protein (FOXO4–DRI–FOXO6), protein–protein (IKBKE–CTNB1), and miRNA–protein (hsa-miR-24-3p–TNF- α) complexes; analyzed interactions on a PBS cluster, identified candidate sites, and guided wet-lab perturbations, with experimental results supporting predictions.
- Designed and prototyped an automated target-exploration framework linking disease text to protein probes, human-proteome cross-validation, and iterative molecular-simulation prioritization.

AI Drug Repurposing – Computational Chemistry Alignment

Ph.D. Core Project | 2022–2024

- Built multi-source retrieval and alignment across PubMed Central, Web of Science, miRTarBase, UniProt, DrugBank, ChEMBL, and PubChem, forming a ~9k-protein $\times$ ~3k-drug search space.
- Applied greedy protein selection and cheminformatics ML with chemical-attribute interpretability, reducing the space to 50 proteins $\times$ 35 drugs.
- Benchmarked general and domain LLMs for biomedical NLP; used GPT for NER and relation extraction across ~4k papers to identify directional disease–protein–drug relationships.
- Implemented an HPC computational-chemistry pipeline (PBS/Slurm) for docking, molecular dynamics, quantum chemistry, and wavefunction analysis; evaluated all 1,750 protein–ligand pairs.
- Prioritized folic acid; validated ranking in HDF and keratinocyte proliferation/migration models, completing an AI–computation–experiment loop.
- Developed a light-curable drug-loaded hydrogel; optimized 3D bioprinting, DMA mechanics, and drug-release assessment for personalized diabetic wound care.

AI Chemical Synthesis Prediction – Computational Augmentation

B.Sc. Core Project | 2019–2021

- Curated and engineered a polyamide nanofiltration membrane dataset, expanding $\sim 10^2$ experimental records into high-dimensional representations for synthesis–performance modelling.
- Developed chemistry-informed augmentation and deep-learning models, expanding $\sim 10^2$ records to $\sim 10^6$ samples; achieved 71.1% flux and 81.6% rejection accuracy.

PUBLICATIONS

- Quantitative computational validation of nanoscale interactions between drug molecules and diabetic wound-related proteins for drug discovery. *ACS Nano Medicine*. 1st author | **2026**
- Deep spatial representation learning of polyamide nanofiltration membranes. *Journal of Membrane Science*. 1st author | **2021**
- Enhancing X-ray sensitization with multifunctional nanoparticles. *Small*. co-corresp. author | **2024**
- Clearance of senescent cells alleviates metabolic dysfunction-associated steatohepatitis. *EASL Congress 2024 Abstract Book / Journal of Hepatology*. 4th author | **2024**
- Rapid SERS inspection of carcinogenic aromatic amines in textiles by using liquid interfacial assembled Au array. *Talanta*. 2nd author | **2021**
- UV-induced room temperature synthesis of microporous ladder polymers with efficient photosensitization. *Reactive and Functional Polymers*. 2nd author | **2019**

MANUSCRIPTS

- 2nd author, revised and resubmitted: Designing 2D membranes for precise separation of molecules with ultra-close molecular weights but different shapes. *Journal of the American Chemical Society*.
- 2nd author, in draft: (1) IKBKE as a β -Catenin kinase regulating mesenchymal stem cell differentiation; and (2) Amlexanox-induced nephrogenic diabetes insipidus independent of IKK ϵ inhibition.

SCHOLARLY ACTIVITIES

- Acknowledged Contributor** Xiang Biao, “Experiencing, education, and the capacity to form experience,” manuscript in scholarly circulation; intended for publication in *Open Times*.
- Peer Reviewer** *Journal of Chemical Theory and Computation* (JIF 6.6); *Frontiers in Chemistry* (JIF 5.5); *Frontiers in Pharmacology* (JIF 6.0); *Frontiers in Immunology* (JIF 8.8). Peak annual JIF, 2021–2025.

SELECTED PATENTS

- A spectral graph fundamental dimension correction and differential analysis method based on moving least squares. *CN109596558B*. 1st inventor, granted **2020**

REGISTERED SOFTWARE COPYRIGHTS

- 5 registered:** (1) AI I-Ching Risk-Decision App Design Software (Reg. No. 16146302); (2) Ziwei Doushu Charting & Risk Decision System (Reg. No. 16412604); (3) Xiaoliuren Charting & Risk Decision System (Reg. No. 16412742); (4) BaZi Charting & Risk Decision System (Reg. No. 16412797); and (5) Hexagram Charting & Risk Decision System (Reg. No. 16713520).
- Original developer and copyright registrant; registered **2025**, rights assigned **2026**.

INTERNSHIPS

- Biomap; Taikang Insurance Group Inc.; Beijing JuCreate Network Technology Co., Ltd. during Ph.D.
- USTC; Technical Institute of Physics and Chemistry (Chinese Academy of Sciences). during B.Sc.

ADDITIONAL INFORMATION

- Languages:** Mandarin Chinese (native); English (full professional proficiency).
- Collaborations:** NUS Computing, Medicine and Additive Manufacturing; CSU Xiangya Medical School; HUST EEE; National Supercomputing Center Changsha.
- Arts:** Member of Hunan Provincial Direct-Affiliated Calligrapher & Artists Association; Hunan Provincial Couplet Association; Changsha Literary Critics Association, etc.; creator of the *Abstract*, *Newton*, and related series; **37** artworks registered for copyright (e.g., Hunan-2025-F-00098719).



LinkedIn



WeChat



ORCID



WhatsApp



GitHub